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MAE 3240

Problem Set #2

Due: 2/10/2026

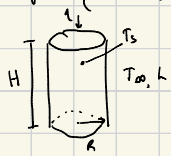
Submitted: 2/10/2026

Acknowledgements:

1) Known: $R = 0.5\,m$, $H = 1\,m$   When full, $T_{in} = 50^\circ C = T_{wall}$, $T_\infty = 20^\circ C$, $h = 15\ W/m^2 K$

Find: a) total power $q$   b) find new $q$ after adding insulation foam

Schematic:




$L = 0.02\,m$ thick insulation, $k = 0.1\ W/mK$

Assumptions: steady state, bottom of water heater insulated, thermal radiation negligible, 1D conduction across insulation foam

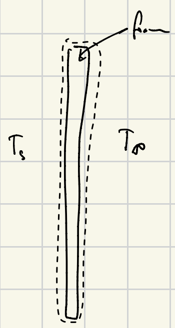
Analysis:

a)
$$q = hA(T_s - T_\infty)$$

$$q = h(\pi R^2 + 2\pi R H)(T_s - T_\infty) = \boxed{1767.15\ W}$$

b) Conduction: $q = -kA \frac{dT}{dx} \Rightarrow q = \frac{-kA\,\Delta T}{L} = -kA\,\frac{T_s - T_{foam}}{L}$

Convection: $q = hA(T_{foam} - T_\infty)$




$E_{in} = E_{out}$

$q_{cond} = q_{convection}$ $\Rightarrow$ $\frac{+kA(T_s - T_{foam})}{L} = hA(T_{foam} - T_\infty)$

$$T_{foam} = 27.5^\circ C$$

$$q = hA(27.5^\circ - 20^\circ C) = \boxed{441.78\ W}$$

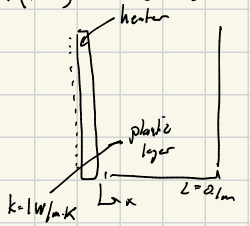
Check: $q = \frac{kA(T_s - T_{foam})}{L} = \frac{0.1 \cdot A(50 - 27.5)}{0.02} = 441.786\ W$

c) The insulating foam decreased the power required by 4 times despite being only 2cm thick. Thus, insulation is largely important to save energy.

2) Known: $L = 10\,cm$, $k = 1\,W/m\cdot K$, $T_\infty = 20^\circ C$, $h$, $100\,W/m^2K$, a) $q'' = 1000\,W/m^2$

Find: a) $T(x=0)$ & $T(x=L)$

Schematic:




Assumptions: steady state, neglect radiation, infinitely thin heater, wall is thermally insulating.

Analysis:

a) $q'' = 1000\,W/m^2$

conduction $q'' = -k\dfrac{dT}{dx} \Rightarrow -k\dfrac{\Delta T}{\Delta x}$

convection $q'' = h(T_s - T_\infty)$


At steady state:

$E_{in} = E_{out}$

$\rightarrow q''_{conduction} = q''_{convection}$

$$1000\,\frac{W}{m^2} = h(T_f - T_\infty) \Longrightarrow T_f = 30\,^\circ C$$

$$1000\,\frac{W}{m^2} = -1\,\frac{W}{mK}\,\frac{(T_s - T_f)}{L} = -1\,\frac{W}{mK}\left(\frac{T_s - 30^\circ C}{0.1}\right)$$

$\rightarrow T_s = 130^\circ C$

$$\boxed{\begin{array}{l} \text{At } x=0 \Rightarrow T = 130^\circ C \\ \text{At } x=L \Rightarrow T = 30^\circ C \end{array}}$$

b) $q_d'' = \frac{1W}{mK}\left(\frac{200 - T_f}{0.1}\right)$

$q_v'' = 100\,(T_f - 20)$

$q_d'' = q_v''$ because $E_{in} = E_{out}$ in CV energy balance at Steady State.

1. $\frac{200 - T_f}{0.1} = 100\,(T_f - 20) \implies T_f = 36.36^\circ C$

$q'' = 100\,(36.36 - 20) = \boxed{1636.16\ W/m^2}$

c) higher $T_f \Rightarrow$ higher $q''$

$$k\left(\frac{200^\circ C - T_f}{L}\right) = h\,(T_f - 20^\circ)$$

$$\frac{200}{L}k - \frac{k}{L}T_f = hT_f - 20h$$

$$\frac{200}{L}k + 20h = hT_f + \frac{k}{L}T_f = \left(h + \frac{k}{L}\right)T_f$$

$$T_f = \frac{\frac{200}{L}k + 20h}{h + \frac{k}{L}}$$

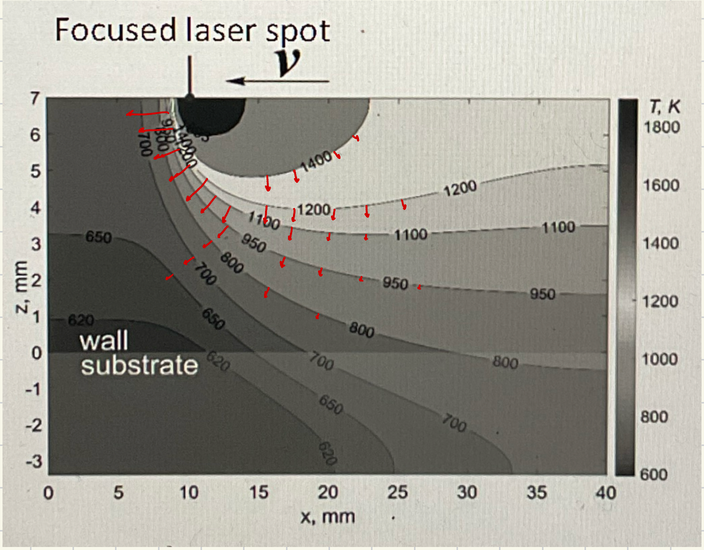
Ways to increase $T_f$: 1. increase $k$ → dissipate more heat

2. increase $h$ → dissipate more heat

3. decrease $L$ → less insulation keeping heat in.

3)

a)




* heat flux vectors normal to contours
* greater magnitude when contours are closer together
* point from greater temperature to lower temperature

b) The contours are much more closely packed on the left side.
This indicates a greater temperature gradient → larger heat flux